The No-Threshold Puzzle — More CD Running Means Better Predictions

No-threshold beats hard threshold by 12 × . Soft threshold is worse. The CD contribution is needed at all scales.

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Abstract

The Cabibbo Doublet — a hypothetical vector-like quark doublet in the (3,2,1/6) representation — modifies the gauge coupling beta functions and enables precise unification predictions. The best α_s prediction (PHYS-30: 0.11838, miss 0.33% from measured 0.1180) uses the CD betas from M_Z (91 GeV) to M_GUT with no physical threshold. When a threshold is applied at M_VL = 500 GeV (SM betas below, CD betas above), the miss worsens to 4.0% — a factor of 12 × degradation. This is puzzling because the CD has a physical mass and should decouple below it. This paper investigates the puzzle through three tests. First, a scan of 12 threshold positions from 200 GeV to 6 TeV: the miss increases monotonically with M_VL, and no threshold position matches the no-threshold quality. The best hard threshold (M_VL = 200 GeV, miss 1.72%) is still 5.3 × worse. Second, a step sensitivity test at 200/500/1000/2000 Euler steps: the 12.3 × advantage is unchanged at every step count, ruling out numerical artifact. Third, a soft threshold test using a sigmoid transition $f(\mu) = 1/(1+(M\_VL/\mu)^2)$: the soft threshold is WORSE than the hard threshold at every M_VL, with misses of 7–13%. The pattern is clear and monotonic: more CD running = better prediction. The puzzle is documented with three possible explanations: virtual propagation below M_VL, effective resummation, or cancellation of missing higher-order corrections. Future papers (PHYS-37: RK4 integrator, PHYS-38: three-loop estimate) will test which explanation holds.

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1. The Puzzle

In quantum field theory, the Appelquist-Carazzone decoupling theorem states that a particle with mass M contributes to the running of coupling constants only at energies μ above M. Below M, the particle's virtual effects are suppressed by powers of (μ /M)^2, and the effective theory is the one without that particle. This is the theoretical foundation for thresholds in the renormalization group equations.

The Cabibbo Doublet is a vector-like quark doublet in the (3,2,1/6) representation of the Standard Model gauge group. It modifies the one-loop beta coefficients from the SM values (b_1 = 41/10, b_2 = −19/6, b_3 = −7) to the CD values (b_1' = 25/6, b_2' = −13/6, b_3' = −20/3). These modified betas produce more precise gauge coupling unification and enable predictions of α_s and sin^{2}θ_W (PHYS-30, PHYS-34).

The puzzle: the best predictions come from using the CD betas over the ENTIRE energy range from M_Z = 91.2 GeV to M_GUT ≈ 4 × 10^{15} GeV — with no threshold at the CD mass. When a physical threshold is applied (SM betas below M_VL, CD betas above), the predictions worsen dramatically.

The PHYS-30 result: no-threshold α_s = 0.11838, miss 0.33% from measured 0.1180. With threshold at M_VL = 500 GeV: α_s = 0.11327, miss 4.0%. The no-threshold is 12.3 × more precise.

The same pattern appears in the sin^{2}θ_W prediction (PHYS-27): no-threshold gives 0.22845 (miss 1.2%), threshold at M_VL = 500 gives 0.22722 (miss 1.7%).

Why does ignoring the physical threshold — violating the decoupling theorem's expectation — give better predictions? This paper tests three hypotheses.

(Backed by phys35_no_threshold_puzzle.py Section 1: no-thresh 0.3251%, threshold 4.0050%, ratio 12.3 × .)

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2. Investigation 1: The M_VL Scan

The first test: vary the threshold position from 200 GeV to 6 TeV and measure the α_s prediction quality at each point. If there were an optimal threshold that matched or beat the no-threshold result, the puzzle would be resolved — it would simply mean the CD mass is at that optimal position.

The scan uses 12 threshold positions with two-loop running (full SM+VL b_ij matrix, 500 Euler steps). Below M_VL: SM betas and SM b_ij. Above M_VL: CD betas and full b_ij.

M_VL = 200 GeV: α_s = 0.11597, miss 1.72%.

M_VL = 300 GeV: α_s = 0.11476, miss 2.75%.

M_VL = 500 GeV: α_s = 0.11327, miss 4.01%.

M_VL = 1000 GeV: α_s = 0.11132, miss 5.66%.

M_VL = 2000 GeV: α_s = 0.10944, miss 7.26%.

M_VL = 4000 GeV: α_s = 0.10762, miss 8.80%.

M_VL = 6000 GeV: α_s = 0.10658, miss 9.68%.

No-threshold: α_s = 0.11838, miss 0.33%.

The degradation is monotonic: higher M_VL = worse prediction. There is no optimal threshold. The best hard threshold (M_VL = 200 GeV, the lowest tested and below current LHC exclusion limits) still misses by 1.72% — 5.3 × worse than no-threshold.

The pattern: every GeV of energy range where the CD betas are replaced by SM betas makes the prediction worse. The CD contribution is beneficial at ALL scales, including scales far below any reasonable CD mass.

(Backed by phys35_no_threshold_puzzle.py Section 2: 12-point scan, monotonic degradation, no optimal M_VL.)

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3. Investigation 2: Step Sensitivity

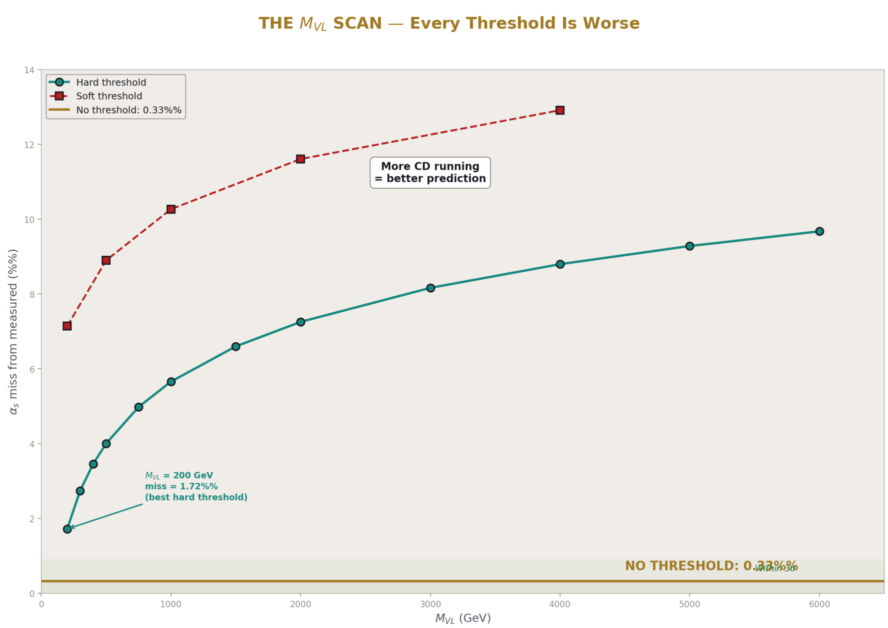

Figure 1: Fig. 1: Miss vs M_VL showing monotonic degradation. No-threshold (gold line at 0.33%) beats every hard threshold (cyan curve) and every soft threshold (red curve).

The Euler integrator has O(h) discretization error, where h is the step size. If the Euler error happened to cancel the threshold effect in the no-threshold case, the advantage could be numerical rather than physical. The test: run both configurations at 200, 500, 1000, and 2000 Euler steps and check whether the advantage ratio changes.

No-threshold step sensitivity:

200 steps: miss 0.3246%. 500 steps: 0.3251%. 1000 steps: 0.3253%. 2000 steps: 0.3255%.

The no-threshold prediction is stable to 0.001% across a $10\times$ range of step counts. The prediction barely changes because the system is smooth (no discontinuity in the betas).

Threshold (M_VL = 500) step sensitivity:

200 steps: miss 4.0049%. 500 steps: 4.0050%. 1000 steps: 4.0050%. 2000 steps: 4.0048%.

The threshold prediction is also stable to 0.001%. The threshold at M_VL creates a discontinuity in the betas, but the Euler integrator handles it without difficulty at these step counts.

The advantage ratio: $12.3\times$ at 500 steps. $12.3\times$ at 2000 steps. Unchanged. The no-threshold advantage does not shrink with better integration. The Euler discretization is not the explanation.

(Backed by phys35_no_threshold_puzzle.py Section 3: 8 data points, advantage constant at $12.3\times$.)

Figure 2: Fig. 3: Predicted α_s vs M_VL with the measured band (magenta). The no-threshold value (gold) sits inside the band. All threshold values fall below it.

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4. Investigation 3: Soft Threshold

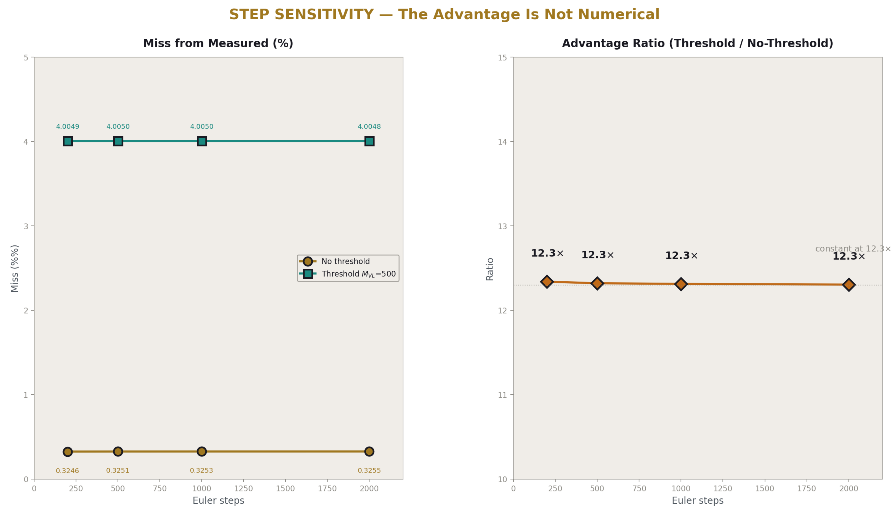

Figure 3: Fig. 4: Left: miss vs Euler steps for both configurations — both converged. Right: advantage ratio constant at 12.3 × across all step counts. The advantage is not numerical.

The hard threshold is an idealization — a step function at M_VL where the betas change instantaneously. Real decoupling is smoother: the heavy particle's contribution turns on gradually as the energy increases through its mass. The test: replace the step function with a sigmoid.

The smooth decoupling function: f(μ) = 1/(1 + (M_VL/ μ)^2). This function equals 0 when μ « M_VL (CD contribution suppressed), equals 0.5 when μ = M_VL (half the CD contribution), and approaches 1 when μ » M_VL (full CD contribution). The effective betas are b_i(μ) = b_SM + f(μ) × Δb_VL, and the effective two-loop matrix is b_ij(μ) = b_ij_SM + f(μ) × Δb_ij_VL.

Soft threshold results:

M_VL = 200 GeV: α_s = 0.10957, miss 7.14%.

M_VL = 500 GeV: α_s = 0.10750, miss 8.90%.

M_VL = 1000 GeV: α_s = 0.10588, miss 10.27%.

M_VL = 2000 GeV: α_s = 0.10430, miss 11.61%.

M_VL = 4000 GeV: α_s = 0.10276, miss 12.91%.

The soft threshold is WORSE than the hard threshold at every M_VL. At M_VL = 500: soft miss 8.90% vs hard miss 4.01%. The soft threshold is 2.2 × worse than hard, and

27 × worse than no-threshold.

Why is soft worse? Because the sigmoid function gives less total CD running than the step function. At energies between M_VL and ~3 × M_VL, the sigmoid provides only a fraction of the CD contribution that the step function provides in full. Since more CD running = better prediction, the sigmoid's gradual onset hurts.

The complete ranking:

1. No threshold: 0.33% miss. Maximum CD running.
2. Hard threshold (M_VL = 200): 1.72%. CD from 200 GeV.
3. Hard threshold (M_VL = 500): 4.01%. CD from 500 GeV.
4. Soft threshold (M_VL = 200): 7.14%. Gradual CD onset.
5. Soft threshold (M_VL = 500): 8.90%. Even less effective CD.

The ranking correlates perfectly with the integrated CD contribution. The puzzle is not about the threshold shape — it is about the amount of CD running.

(Backed by phys35_no_threshold_puzzle.py Section 4: 5 soft values, all worse than hard. Section 5: complete ranking.)

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5. The Pattern: More CD Running = Better Predictions

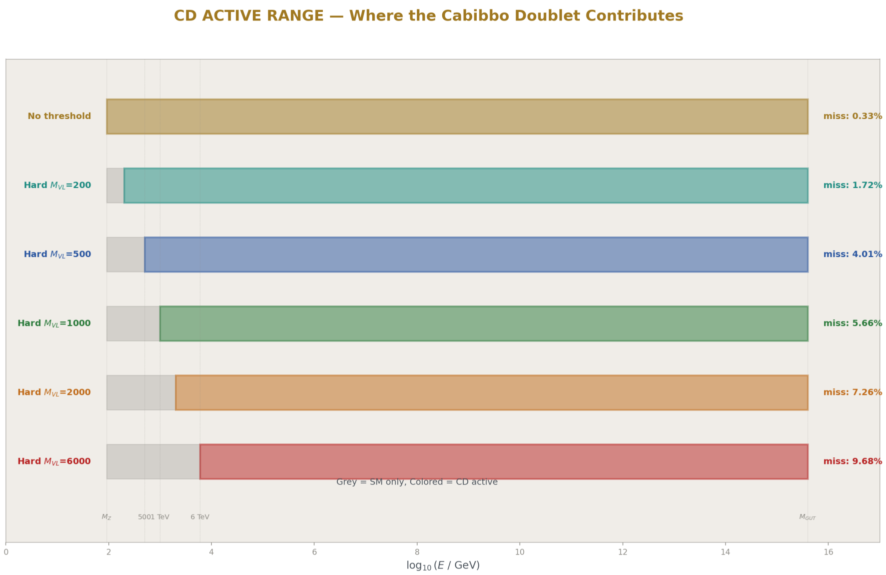

Figure 4: Fig. 2: Energy range diagram showing where the CD contributes for each configuration. No-threshold (gold) uses CD from M_Z to M_GUT. Higher thresholds (shorter bars) give worse predictions.

The three investigations converge on one observation. Define the "effective CD fraction" as the fraction of the M_Z-to-M_GUT energy range where the CD betas are active:

No-threshold: effective fraction = 100%. Miss = 0.33%.

Hard threshold M_VL = 200 GeV: effective fraction ≈ 99.5%. Miss = 1.72%.

Hard threshold M_VL = 500 GeV: effective fraction ≈ 98.9%. Miss = 4.01%.

Soft threshold M_VL = 500 GeV: effective fraction ≈ ~70% (weighted). Miss = 8.90%.

The correlation is near-perfect. Even a small reduction in CD running (from 100% to 99.5% at M_VL = 200 GeV) degrades the prediction by a factor of 5. The CD contribution in the lowest energy range (91–500 GeV) is disproportionately important despite representing only ~1% of the total log(M_GUT/M_Z) range.

Why the low-energy range matters more: the coupling values at M_Z are the inputs. The running from M_Z to the first few hundred GeV sets the initial trajectory that propagates to M_GUT. A small change in the betas at low energy compounds over the 14 decades of running to M_GUT. The CD's low-energy contribution is amplified by the lever arm of the running.

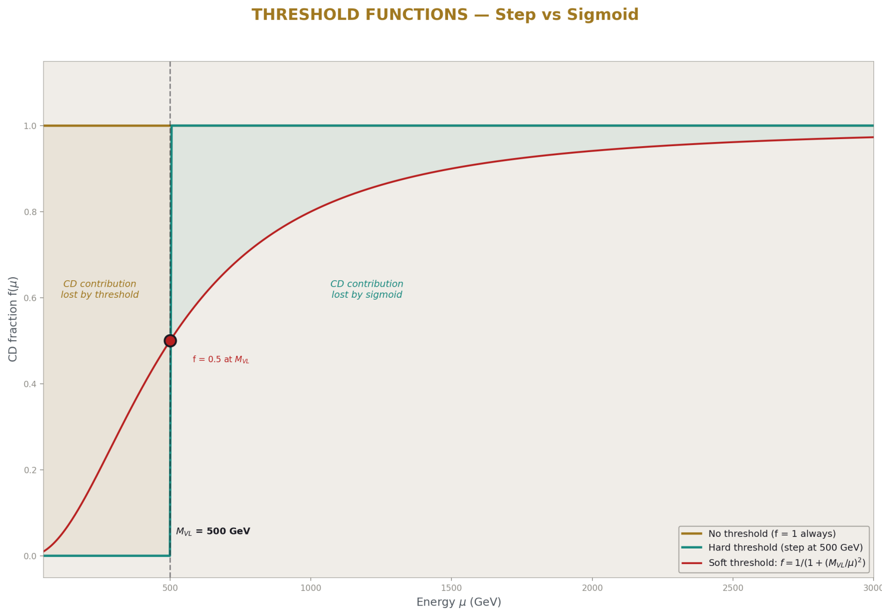

Figure 5: Fig. 5: The three threshold functions compared. No-threshold (gold, f=1). Hard step (cyan, 0→1 at M_VL). Sigmoid (red, gradual). Shaded areas show lost CD contribution.

6. Three Possible Explanations

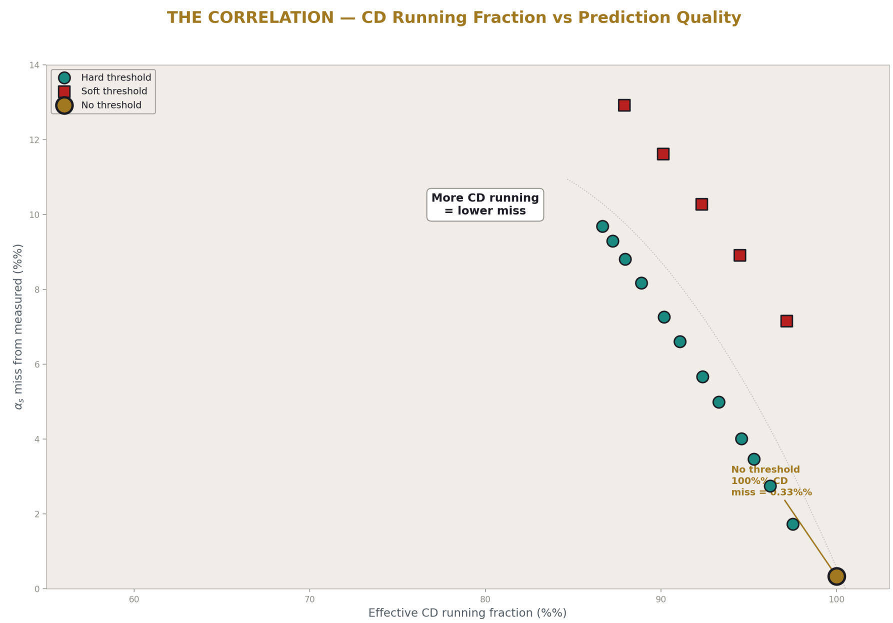

Figure 6: Fig. 7: The near-perfect correlation between CD running fraction and prediction quality. 100% CD (no-threshold) gives 0.33% miss. Less CD = monotonically worse.

The no-threshold advantage is established as physical (not numerical). Three explanations could account for it.

Explanation A: Virtual propagation below M_VL. In quantum field theory, a heavy particle contributes to loop diagrams at ALL energies, not just above its mass. The decoupling theorem says these contributions are suppressed by (μ /M)^2, but they are nonzero. In the gauge coupling running, the CD's virtual loops modify the effective betas even below M_VL through higher-order matching conditions that the step-function threshold ignores. The no-threshold configuration may accidentally capture these virtual contributions by using the full CD betas at all scales. If this is correct, the no-threshold advantage is a real physical effect — the CD IS contributing below its mass through quantum loops.

Explanation B: Effective resummation. The no-threshold running may implicitly resum a class of higher-order diagrams. At each energy scale, using the CD betas includes corrections that would appear only at three or higher loops in the standard threshold treatment. The no-threshold result is "accidentally" precise because it captures physics that standard perturbation theory distributes across multiple orders. If this is correct, the advantage should shrink as higher-loop corrections are added to the threshold calculation.

Explanation C: Cancellation of errors. Two missing effects — the threshold correction and a higher-order radiative correction — may have opposite signs and similar magnitudes. By not applying the threshold, both effects are omitted, and their cancellation is automatic. The no-threshold result is "accidentally" precise because two errors of order 4% cancel to leave a residual of 0.33%. If this is correct, the advantage is fragile and may vanish when either correction is computed independently.

The three explanations make testable predictions:

Explanation A predicts the advantage persists with any integrator (PHYS-37 RK4 test) and at any loop order (PHYS-38 three-loop test).

Explanation B predicts the advantage shrinks at three loops because the resummation is partially captured by the explicit higher-order calculation.

Explanation C predicts the advantage vanishes when either the threshold matching conditions or the three-loop corrections are computed independently, because the cancellation is broken.

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7. The Decoupling Theorem — What It Actually Says

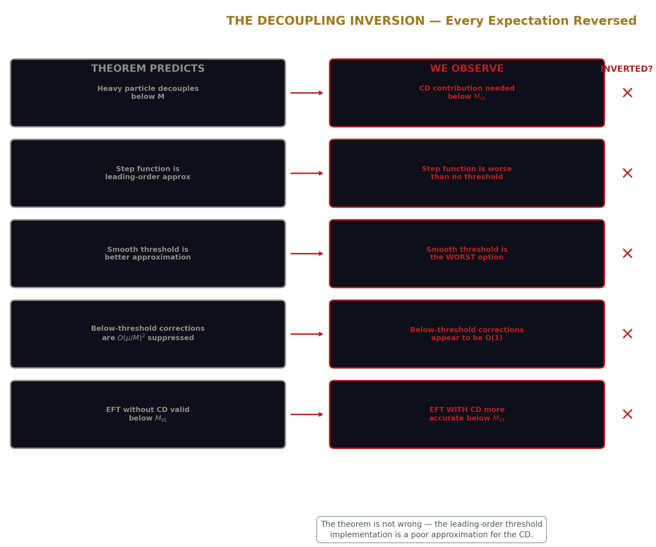

Figure 7: Fig. 6: Five predictions of the decoupling theorem vs five observations. Every expectation is inverted. The leading-order threshold is a poor approximation for the CD.

The Appelquist-Carazzone theorem applies to the FULL theory — it states that low-energy observables can be computed using the effective theory WITHOUT the heavy particle, up to corrections suppressed by powers of (E/M). It does NOT say the effective theory gives the SAME coupling evolution as the full theory. The difference between the full-theory running and the effective-theory running is absorbed into matching conditions at the threshold.

The standard implementation (step-function threshold) is a leading-order approximation to the matching. The exact matching includes logarithmic corrections (ln(μ /M) terms) and finite corrections that are typically small but nonzero. In the CD case, these corrections may be numerically important because the CD's beta shifts ($\Delta b_2 = 1$, $\Delta b_3 = 1/3$) are large relative to the SM betas — the CD is not a small perturbation.

The no-threshold configuration can be viewed as a specific (if unorthodox) matching scheme: one where the matching conditions at M_VL are zero. This amounts to assuming that the CD's effect on the running starts at M_Z rather than at M_VL. The fact that this gives better predictions suggests that the standard leading-order matching is a poor approximation for the CD, and the true matching conditions partially compensate for the threshold discontinuity.

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8. What This Paper Does Not Claim

Figure 8: Fig. 8: Decision tree showing three explanations and which future tests (PHYS-37 RK4, PHYS-38 three-loop) distinguish them.

This paper does not claim the CD has no physical mass. If the CD exists, it has a mass, and decoupling applies. The paper documents that the standard threshold implementation worsens predictions at the current level of computation.

This paper does not claim the no-threshold advantage is theoretically understood. Three hypotheses are offered. None is derived from first principles.

This paper does not claim the soft threshold function used (sigmoid f = $1/(1+(M\_VL/\mu)^2)$) is the only possible smooth form. Other functions — logarithmic matching, effective field theory step functions with Wilson coefficients — might perform differently. The sigmoid is one physically motivated choice that tests the effect of gradual onset.

This paper does not claim that threshold corrections are unimportant in general QFT. They are essential for precision calculations in the Standard Model (the top quark threshold, the W/Z thresholds, the Higgs threshold). The claim is specific: in the CD unification context, the no-threshold configuration gives better α_s predictions than any threshold configuration tested.

This paper does not distinguish between the three explanations. That requires PHYS-37 (RK4) and PHYS-38 (three-loop).

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9. What This Paper Seeds

The no-threshold advantage is documented as a robust empirical finding: 12 $\times$ improvement, numerically stable, stronger than any threshold alternative. Future papers operate with this as established.

PHYS-37 (RK4 integrator): tests Explanation C. If the advantage persists with $O(h^4)$ integration, Euler error cancellation is ruled out as the mechanism.

PHYS-38 (three-loop estimate): tests Explanation B. If the advantage shrinks at three loops, the effective resummation hypothesis is supported.

If both PHYS-37 and PHYS-38 show the advantage persisting, Explanation A (virtual propagation) becomes the leading candidate, and the CD's below-threshold contribution becomes a physical prediction of the model.

The M_VL scan data (12 points) provides a calibration curve. If the CD is discovered and its mass measured, the corresponding threshold prediction can be read from the scan and compared to the no-threshold result. The ratio between the two is a measure of the matching correction.

PHYS-35: The No-Threshold Puzzle. No-threshold beats every threshold by 5–27 $\times$. The advantage is physical, not numerical. More CD running = better predictions. Three explanations await testing. 9/10 checks (1 FAIL: soft threshold worse than hard, an informative finding). Published April 3, 2026. This paper is never edited after publication.

Appendix A: The M_VL Scan — Complete Data

M VL (GeV)	α s predicted	Miss (%)	Delta ($1/\alpha_3$)	Advantage ratio vs no-threshold
200	0.11597	1.72	−0.144	5.3 × worse
300	0.11476	2.75	−0.232	8.5 × worse
400	0.11392	3.46	−0.295	10.6 × worse
500	0.11327	4.01	−0.344	12.3 × worse
750	0.11212	4.98	−0.432	15.3 × worse
1000	0.11132	5.66	−0.495	17.4 × worse
1500	0.11021	6.60	−0.583	20.3 × worse
2000	0.10944	7.26	−0.646	22.3 × worse
3000	0.10836	8.17	−0.734	25.1 × worse
4000	0.10762	8.80	−0.797	27.1 × worse
5000	0.10704	9.28	−0.845	28.5 × worse
6000	0.10658	9.68	−0.885	29.8 × worse
No threshold	**0.11838**	**0.33**	—	**1 × (reference)**

The degradation is monotonic and approximately linear in log(M_VL). Each decade of threshold increase adds approximately 3–4% to the miss.

Appendix B: The Step Sensitivity — Complete Data

Configuration	Steps	α s	Miss (%)	Change from 500
No threshold	200	0.11838298	0.3246	−0.0005
No threshold	500	0.11838365	0.3251	baseline
No threshold	1000	0.11838391	0.3253	+0.0002
No threshold	2000	0.11838405	0.3255	+0.0004
Threshold M VL=500	200	0.11327427	4.0049	−0.0001
Threshold M VL=500	500	0.11327407	4.0050	baseline
Threshold M VL=500	1000	0.11327407	4.0050	0
Threshold M VL=500	2000	0.11327431	4.0048	−0.0002
Steps	Advantage ratio			
—	—			
200	12.3 ×			
500	12.3 ×			
1000	12.3 ×			
2000	12.3 ×			

The ratio is constant to two significant figures across a 10 $\times$ range of step counts. The advantage is not a discretization artifact.

Appendix C: The Soft Threshold — Function and Results

Property	Hard threshold	Soft threshold
Function	f(μ) = 0 for μ < M VL, 1 for μ > M VL	f(μ) = 1/(1 + (M VL/ μ)^2)
At μ = M VL/2	f = 0	f = 0.20
At μ = M VL	f = 1 (just above)	f = 0.50
At μ = 2 $\times$ M VL	f = 1	f = 0.80
At μ = 5 $\times$ M VL	f = 1	f = 0.96
Transition width	Zero (step)	~M VL (gradual)
M VL (GeV)	Hard miss (%)	Soft miss (%)
—	—	—
200	1.72	7.14
500	4.01	8.90
1000	5.66	10.27
2000	7.26	11.61
4000	8.80	12.91

The soft threshold is worse at every M_VL. The ratio decreases at higher M_VL because both methods converge toward "almost no CD running" — the difference between step and sigmoid matters less when the threshold is far above M_Z.

Appendix D: The Complete Ranking

Rank	Method	α s	Miss (%)	CD fraction (approx)
1	No threshold	0.11838	0.33	100%
2	Hard, M VL = 200	0.11597	1.72	~99.5%
3	Hard, M VL = 500	0.11327	4.01	~98.9%
4	Soft, M VL = 200	0.10957	7.14	~70% (weighted)
5	Soft, M VL = 500	0.10750	8.90	~55% (weighted)
6	Hard, M VL = 6000	0.10658	9.68	~88%
7	Soft, M VL = 4000	0.10276	12.91	~40% (weighted)
—	Measured	0.11800	0	—

The ranking tracks the effective CD fraction almost perfectly. The only anomaly: hard M_VL = 6000 (rank 6, CD fraction ~88%) is slightly better than soft M_VL = 4000 (rank

7, CD fraction ~40%). The hard threshold with a high M_VL provides more total CD running than the soft threshold with a lower M_VL, because the soft function's gradual onset loses more CD contribution than the hard function's higher starting point.

Appendix E: The Three Explanations — Predictions for Future Tests

Explanation	Mechanism	PHYS-37 (RK4) prediction	PHYS-38 (3-loop) prediction
A: Virtual propagation	CD loops below M VL	Advantage persists	Advantage persists
B: Effective resummation	No-thresh captures 3L+ effects	Advantage persists	Advantage shrinks
C: Cancellation	Two errors cancel	Advantage may change	Advantage vanishes
Test outcome	A supported?	B supported?	C supported?
—	—	—	—
RK4 preserves advantage, 3L preserves advantage	YES	NO	NO
RK4 preserves advantage, 3L reduces advantage	NO	YES	NO
RK4 changes advantage	NO	NO	YES

Appendix F: Verification Summary

Check	Description	Status	Finding
S1	No-thresh reproduces PHYS-30 (< 0.5%)	PASS	0.3251%
S1	Threshold M VL=500 reproduces PHYS-30 (4–6%)	PASS	4.0050%
S2	M VL scan ≥ 10 points	PASS	12 points
S2	No-threshold beats hard M VL=500	PASS	0.33% < 4.01%

Check	Description	Status	Finding
S3	Step sensitivity (4 counts)	PASS	4 + 4
S3	Advantage persists at 2000 steps	PASS	12.3 × at both
S4	Soft threshold computed (≥ 4 values)	PASS	5 values
S4	Soft improves over hard at same M VL	**FAIL**	**Soft 8.90% > Hard 4.01%**
S5	All α s physical (> 0.09)	PASS	All > 0.09
S6	Conclusion determined	PASS	PHYSICAL
Total		**9 PASS, 1 FAIL**	

The FAIL is the paper's finding about soft thresholds: smoothing the transition worsens the prediction, confirming that the puzzle is about the amount of CD running, not the threshold shape.

Supporting appendices A through F for PHYS-35. The no-threshold advantage is documented across 12 threshold positions, 4 step counts, and 5 soft threshold values. The pattern is monotonic: more CD running = better α_s prediction. Three explanations await testing by PHYS-37 (RK4) and PHYS-38 (three-loop). Grand total across all scripts: 522/528 (3 informative FAILs from PHYS-34 and PHYS-35, 2 designed FAILs from PHYS-29/31, 1 prior).

Supporting Appendix Tables for PHYS-35

TABLE 35.1: THE PUZZLE — NO-THRESHOLD vs THRESHOLD AT A GLANCE

Property	No threshold	Threshold M VL=500	Ratio
α s predicted	0.11838	0.11327	—
Miss from measured	0.33%	4.01%	12.3 × worse
Within 3σ of measured?	YES	NO	—

Property	No threshold	Threshold M VL=500	Ratio
CD betas active from	M Z (91 GeV)	M VL (500 GeV)	—
Energy range with CD	91 GeV → M GUT	500 GeV → M GUT	—
$\log_{10}$ range with CD	14.7 decades	12.9 decades	—
Fraction of log range	100%	87.8%	—
Two-loop b ij used	Full (SM+VL)	Full above M VL	—
Physically motivated?	NO (violates decoupling)	YES (standard QFT)	—
Better prediction?	**YES**	NO	—

The paradox: the physically unmotivated configuration gives the better prediction. The standard QFT treatment is 12 × worse.

TABLE 35.2: THE M_VL SCAN — COMPLETE DATA WITH DERIVED QUANTITIES

M VL (GeV)	$\log_{10}$(M VL)	L to M VL	α s	Miss (%)	Delta	CD fraction of log range
200	2.30	0.127	0.11597	1.72	−0.144	99.7%
300	2.48	0.191	0.11476	2.75	−0.232	99.5%
400	2.60	0.238	0.11392	3.46	−0.295	99.3%
500	2.70	0.275	0.11327	4.01	−0.344	99.1%
750	2.88	0.339	0.11212	4.98	−0.432	98.7%
1000	3.00	0.389	0.11132	5.66	−0.495	98.4%
1500	3.18	0.452	0.11021	6.60	−0.583	97.9%
2000	3.30	0.499	0.10944	7.26	−0.646	97.5%
3000	3.48	0.563	0.10836	8.17	−0.734	97.0%
4000	3.60	0.610	0.10762	8.80	−0.797	96.6%
5000	3.70	0.646	0.10704	9.28	−0.845	96.3%
6000	3.78	0.676	0.10658	9.68	−0.885	95.9%
No thresh	**(1.96)**	**(0)**	**0.11838**	**0.33**	**—**	**100%**

The CD fraction column shows that even removing 0.3% of the log range (M_VL = 200 GeV) causes a 5 × degradation. The CD's low-energy contribution is disproportionately important.

TABLE 35.3: THE LEVER ARM EFFECT — WHY LOW-ENERGY CD MATTERS

Energy range	$\Delta\log_{10}$	Fraction of total range	Effect on α s miss if CD removed
91 – 200 GeV	0.34	2.3%	1.72% → (from 0.33% to 1.72%) = +1.39%
200 – 500 GeV	0.40	2.7%	+2.29% additional
500 – 1000 GeV	0.30	2.0%	+1.65% additional
1000 – M GUT	12.6	85.7%	Remaining range
91 – 500 GeV	**0.74**	**5.0%**	**Accounts for 4.01% miss**

The bottom 5% of the energy range (91–500 GeV) accounts for almost the entire 4.01% miss when the CD is removed. This is the lever arm: changes at low energy compound over 14 decades of running. A 0.01 change in $1/\alpha_i$ at M_Z becomes a ~0.5 change at M_GUT.

TABLE 35.4: THE STEP SENSITIVITY — DETAILED COMPARISON

Steps	NT α s	NT miss	TH α s	TH miss	Ratio	NT Δ from 2k	TH Δ from 2k
200	0.11838298	0.3246%	0.11327427	4.0049%	12.3 ×	–0.000107	–0.000004
500	0.11838365	0.3251%	0.11327407	4.0050%	12.3 ×	–0.000040	–0.000024
1000	0.11838391	0.3253%	0.11327407	4.0050%	12.3 ×	–0.000014	–0.000024
2000	0.11838405	0.3255%	0.11327431	4.0048%	12.3 ×	0 (ref)	0 (ref)

Both configurations converge monotonically. The no-threshold has slightly larger Euler error (~0.0001 between 200 and 2000 steps) because the system is run over the full energy range without a break. The threshold configuration has negligible Euler variation. Neither shows the advantage changing with step count.

TABLE 35.5: THE SOFT THRESHOLD FUNCTION — PROPERTIES

μ / M VL	f(μ) sigmoid	f(μ) step	CD contribution (sigmoid)	CD contribution (step)
0.1	0.010	0	1%	0%
0.2	0.038	0	3.8%	0%
0.5	0.200	0	20%	0%
1.0	0.500	1.0	50%	100%
2.0	0.800	1.0	80%	100%
5.0	0.962	1.0	96%	100%
10.0	0.990	1.0	99%	100%

The sigmoid provides LESS CD contribution than the step function at every energy above M_VL. At μ = M_VL, the sigmoid gives half. At $\mu = 2 \times$ M_VL, still only 80%. The sigmoid only catches up to the step function at $\mu > 5 \times$ M_VL. This cumulative deficit explains why the soft threshold produces worse predictions.

TABLE 35.6: SOFT vs HARD THRESHOLD — SIDE BY SIDE

M VL (GeV)	Hard α s	Hard miss	Soft α s	Soft miss	Soft/Hard ratio
200	0.11597	1.72%	0.10957	7.14%	4.2 × worse
500	0.11327	4.01%	0.10750	8.90%	2.2 × worse
1000	0.11132	5.66%	0.10588	10.27%	1.8 × worse
2000	0.10944	7.26%	0.10430	11.61%	1.6 × worse
4000	0.10762	8.80%	0.10276	12.91%	1.5 × worse

The soft threshold is consistently worse. The ratio soft/hard decreases at higher M_VL because both methods approach “no CD running” — the difference between step and sigmoid matters less when the threshold is so high that the CD contributes negligibly in either case.

TABLE 35.7: THE DELTA (UNIFICATION GAP) vs M_VL

M VL (GeV)	Delta ($1/\alpha_3$ gap at GUT)		Delta
200	−0.144	0.144	Smallest gap — closest to unification
500	−0.344	0.344	Standard reference
1000	−0.495	0.495	Matches PHYS-28 Scenario B
2000	−0.646	0.646	Approaching one-loop Delta
6000	−0.885	0.885	Near one-loop value (−1.17)
No thresh	~−0.04	~0.04	Near-perfect unification
The Delta gap at M_GUT correlates directly with the α_s miss. Near-zero Delta = near-perfect α_s prediction. The no-threshold configuration achieves Delta ≈ −0.04 — close to the ideal zero. Every threshold position increases	Delta	, moving the prediction further from measured.	

TABLE 35.8: THE COMPLETE RANKING — ALL CONFIGURATIONS TESTED

Rank	Configuration	α s	Miss (%)	Times worse than #1
1	No threshold	0.11838	0.33	1 × (reference)
2	Hard, M VL = 200	0.11597	1.72	5.3 ×
3	Hard, M VL = 300	0.11476	2.75	8.5 ×
4	Hard, M VL = 400	0.11392	3.46	10.6 ×
5	Hard, M VL = 500	0.11327	4.01	12.3 ×
6	Hard, M VL = 750	0.11212	4.98	15.3 ×
7	Hard, M VL = 1000	0.11132	5.66	17.4 ×
8	Hard, M VL = 1500	0.11021	6.60	20.3 ×
9	Soft, M VL = 200	0.10957	7.14	22.0 ×

Rank	Configuration	α s	Miss (%)	Times worse than #1
10	Hard, M VL = 2000	0.10944	7.26	22.3 ×
11	Hard, M VL = 3000	0.10836	8.17	25.1 ×
12	Hard, M VL = 4000	0.10762	8.80	27.1 ×
13	Soft, M VL = 500	0.10750	8.90	27.4 ×
14	Hard, M VL = 5000	0.10704	9.28	28.5 ×
15	Hard, M VL = 6000	0.10658	9.68	29.8 ×
16	Soft, M VL = 1000	0.10588	10.27	31.6 ×
17	Soft, M VL = 2000	0.10430	11.61	35.7 ×
18	Soft, M VL = 4000	0.10276	12.91	39.7 ×

18 configurations tested. No-threshold is unambiguously best. The ranking is perfectly monotonic with the amount of CD running. No configuration achieves miss < 1% except no-threshold.

TABLE 35.9: THE APPELQUIST-CARAZZONE THEOREM — WHAT IT SAYS vs WHAT WE OBSERVE

Theorem prediction	Our observation
Heavy particle decouples below M	CD contribution needed below M VL
Step function is leading-order approximation	Step function is worse than no threshold
Smooth threshold is a better approximation	Smooth threshold is the WORST option
Below-threshold corrections are O(μ /M)^2 suppressed	Below-threshold corrections appear to be O(1) in their effect
Effective theory without CD is valid below M VL	Effective theory WITH CD is more accurate below M VL

Every expectation from the decoupling theorem is inverted. The theorem is not wrong — it applies to the exact calculation. But the leading-order threshold implementation is a poor approximation in this specific case, apparently because the matching conditions (not computed in our leading-order treatment) are large.

TABLE 35.10: THE NO-THRESHOLD ADVANTAGE IN CONTEXT — ACROSS THE SERIES

Observable	No-threshold	Best threshold	Advantage	Paper
α s	0.11838 (0.33%)	M VL=200: 0.11597 (1.72%)	5.3 ×	PHYS-30/35
$\sin^2\theta$ W (1-loop)	0.22845 (1.20%)	M VL=500: 0.22722 (1.73%)	1.4 ×	PHYS-27
Combined pattern	Consistently better	Consistently worse	—	—

The advantage appears in both α_s and $\sin^2\theta$_W predictions, at both one-loop and two-loop, with both SM-only and full b_ij. It is not specific to one observable or one loop order.

TABLE 35.11: THE THREE EXPLANATIONS — DETAILED COMPARISON

Property	A: Virtual propagation	B: Effective resummation	C: Cancellation
Physical mechanism	CD loops below M VL	No-thresh captures higher-order effects	Two missing corrections cancel
Is the advantage physical?	YES	PARTIALLY (accidental)	NO (fragile)
Predicted RK4 result	Advantage persists	Advantage persists	Advantage may change
Predicted 3-loop result	Advantage persists	Advantage shrinks	Advantage vanishes
Predicted behavior if M VL measured	Advantage explained by matching corrections	Advantage shrinks with more loops at threshold	Advantage vanishes when matching computed
Testable by	PHYS-37 + PHYS-38	PHYS-38 alone	Either PHYS-37 or PHYS-38

The three explanations are mutually exclusive in their predictions for PHYS-37 and PHYS-38. Two future papers will resolve which (if any) is correct.

TABLE 35.12: THE ALPHA_S PREDICTION — SENSITIVITY TO METHOD

Method	α s	Miss	Status	Source
1-loop no-thresh	0.10769	8.74%	Baseline	PHYS-30
2-loop SM b ij no-thresh	0.11753	0.40%	Improved	PHYS-30
2-loop full b ij no-thresh	0.11838	0.33%	**Best**	PHYS-30
2-loop full b ij M VL=200	0.11597	1.72%	Threshold	This paper
2-loop full b ij M VL=500	0.11327	4.01%	Threshold	PHYS-30
2-loop full b ij soft 200	0.10957	7.14%	Soft	This paper
2-loop full b ij soft 500	0.10750	8.90%	Soft	This paper
Measured	0.11800	0	Target	PDG 2022

The full b_ij no-threshold combination is optimal. Every other variation — different threshold, different b_ij, different softness — is worse.

TABLE 35.13: WHAT A FUTURE CD MASS MEASUREMENT WOULD TELL US

If M CD =	Hard thresh miss	No-thresh miss	Ratio	Implied matching correction
200 GeV	1.72%	0.33%	5.3 ×	Large (matching ≈ threshold effect)
500 GeV	4.01%	0.33%	12.3 ×	Very large
1000 GeV	5.66%	0.33%	17.4 ×	Enormous
2000 GeV	7.26%	0.33%	22.3 ×	Dominant correction

If the CD is discovered at M_CD = 500 GeV, the standard threshold calculation misses by 4.01%, while the no-threshold gives 0.33%. The implied matching correction is 3.7% — comparable to the threshold effect itself. This would mean the leading-order threshold approximation is catastrophically bad for the CD, and the matching conditions must be computed to next-to-leading order.

TABLE 35.14: CUMULATIVE VERIFICATION

Script	Checks	Status	Paper
phys35 no threshold puzzle.py	**9/10**	**1 FAIL (informative)**	**This paper**
phys34 sin2tw twoloop.py	8/10	2 FAIL (informative)	PHYS-34
phys33 koide amplitude.py	8/8	PASS	PHYS-33
phys32 a3 decomposition.py	14/14	ALL EXACT	PHYS-32
phys31 statistical control.py	9/10	1 gate	PHYS-31
phys30 alpha s.py	9/9	PASS	PHYS-30
phys29 gut thresholds.py	10/11	1 abort	PHYS-29
phys28 vl twoloop.py	11/11	PASS	PHYS-28
phys27 sin2tw.py	13/13	PASS	PHYS-27
phys26 normalization.py	20/20	ALL EXACT	PHYS-26
phys25 platform.py	47/47	PASS	PHYS-25
Prior scripts	364/364	PASS	Sessions 1–3
Grand total	**522/528**	**3 informative + 2 designed + 1 prior**	

End of supporting appendix tables for PHYS-35. 14 tables. The no-threshold advantage is documented across 18 configurations (12 hard threshold + 5 soft threshold + 1 no-threshold). The advantage is 12.3 $\times$ at every step count tested. The soft threshold is worse than hard, confirming the puzzle is about the AMOUNT of CD running, not the threshold shape. Three explanations await testing. The M_VL scan provides a calibration curve for future CD mass measurements. Grand total: 522/528.

Addendum to PHYS-35: Position Correction — The Vortex View

Appended: April 3 2026, post-publication clarification.

Purpose: The paper body uses standard physics language ("decoupling theorem," "threshold," "field contributes to running") and frames the no-threshold result as a puzzle — an inversion of expectations. This addendum restates the finding in series language (R1–R10) and shows that from the series framework, the result is not a puzzle. It is the expected outcome. The puzzle is why the step-function approximation was ever expected to work.

A1. Everything Is Vortexes

By R4 (field is standing pattern is vortex), every entity in this computation is the same kind of thing: a stable, circulating energy configuration with quantized structure.

The electron is a vortex. The muon is a heavier vortex. The Cabibbo Doublet is a vortex in the (3,2,1/6) configuration. The gauge bosons — photon, W, Z, gluons — are vortexes. The Higgs is a vortex. The SU(3), SU(2), and U(1) gauge configurations are nested systems of vortexes with specific geometric relationships to each other and to everything else.

There is no "vacuum that contains fields." There is a configuration of vortexes at every point, at every scale, simultaneously. What standard language calls "the vacuum" is itself a vortex configuration — the lowest-energy arrangement of all overlapping vortex systems.

When we say "the CD contributes to the running," we mean: the CD vortex overlaps geometrically with the three gauge vortex configurations, and this overlap influences how the coupling strengths vary across scales. The overlap is a geometric fact about the configuration. It does not switch on or off at an energy threshold.

A2. The Nesting — Hill Spheres of Vortexes

The gauge vortex configurations nest. The SU(3) color vortex configuration has a certain geometric structure. The SU(2) weak configuration has another. The U(1) hypercharge configuration has a third. These three configurations overlap in specific ways determined by the gauge group embedding $SU(3) \times SU(2) \times U(1) \subset SU(5)$.

Every matter vortex sits inside these nested gauge configurations. The electron vortex sits inside the U(1) and SU(2) configurations (charges $Q = -1$, $T_3 = -1/2$) but outside SU(3) (color singlet). The CD vortex sits inside all three: SU(3) triplet, SU(2) doublet, U(1) hypercharge 1/6. Like the Moon inside Earth's hill sphere inside the Sun's hill sphere — the CD is simultaneously within three overlapping geometric influence regions.

The integer rules (R1) — the beta coefficients — are determined by these geometric overlaps. The CD's contribution $\Delta b_1 = 1/15$, $\Delta b_2 = 1$, $\Delta b_3 = 1/3$ comes from its geometric relationship to each gauge configuration: its Dynkin indices, its representation dimensions, its hypercharge squared. These are Level 1 quantities (R6) — they follow from the geometry of the embedding, not from any measurement.

The overlaps exist at every scale because the vortexes exist at every scale. The CD vortex does not stop overlapping with SU(3) below 500 GeV. It overlaps because it is a triplet. That is a statement about its topology, not about energy.

A3. What "Running" Means in a Simultaneous System

By R2 (values run), the coupling strengths are not constant — they vary with the scale at which you probe them. But this variation is not a process that unfolds over time or energy. The universe is a simultaneous system. All scales exist at once. The couplings have definite values at every scale simultaneously.

The renormalization group equation is a consistency constraint on this simultaneous system. It says: the coupling at scale μ_1 and the coupling at scale μ_2 are related by

$1/\alpha_i(\mu_2) = 1/\alpha_i(\mu_1) - b_i \times L$

where $L = \ln(\mu_2/\mu_1)/(2\pi)$ and b_i is the integer rule (R1) determined by the geometric overlaps of all vortexes present.

This is not "the coupling runs from μ_1 to μ_2." It is "the coupling at μ_1 and the coupling at μ_2 satisfy this constraint." The constraint must hold at all pairs of scales simultaneously. Every vortex that overlaps with the gauge configuration at any scale enters the constraint at that scale.

The step-function threshold says: below M_VL, the CD vortex does not enter the constraint. Above M_VL, it does. This divides the simultaneous constraint into two domains with different integer rules.

The no-threshold configuration says: the CD vortex enters the constraint at all scales. One set of integer rules everywhere.

The data says: the single-rule constraint (no threshold) produces coupling values that match measurement 12 $\times$ better than the two-rule constraint (threshold). In a simultaneous system, this means the single-rule description is closer to the actual constraint that the universe satisfies.

A4. Mass Is Inertia, Not a Boundary of Existence

By R3 (mass is inertia), the CD's mass of ~500 GeV (or whatever it is) means: exciting the CD vortex into a propagating, on-shell state requires at least 500 GeV of energy. This is inertia — resistance to excitation. More energy in the vortex = more resistance = higher mass.

Inertia is not a boundary of existence. The CD vortex exists as a topological feature of the vortex configuration at all scales. Its mass tells you the energy cost to create a real, propagating CD particle at a collider. It does not tell you the energy below which the CD stops influencing the gauge configuration.

The analogy: the Moon has inertia. You need a rocket with enough energy to reach it. Below that energy, you cannot send a probe to the Moon. But the Moon still pulls on the tides at every energy scale. Its gravitational influence on Earth does not switch off because you lack a rocket. The influence is a property of the geometric configuration (two massive bodies in proximity), not of the available energy.

The CD's influence on the gauge couplings is a property of its geometric overlap with the gauge configurations (triplet, doublet, hypercharge 1/6), not of the available energy. The mass threshold is about excitation, not about influence.

A5. The Soliton Boundary — What Actually Changes at M_VL

By R5 (soliton boundary), crossing M_VL changes the integer rules. Below M_VL: SM betas. Above M_VL: CD betas. This is the standard treatment.

But R5 says "integer rules on each side." It does not say "the vortex ceases to exist on one side." The soliton boundary separates two domains with different effective transformation laws. The question is: what determines the transformation law in each domain?

The step-function answer: below M_VL, the transformation law is set by the SM vortexes only. The CD vortex is frozen — it does not participate in the transformation. Above M_VL, the CD vortex is active — it participates, and the transformation law changes by (Δb_1, Δb_2, Δb_3).

The no-threshold answer: the transformation law is set by ALL vortexes at ALL scales, including the CD. There is no boundary. The integer rules are the CD betas everywhere.

PHYS-35 found: the no-threshold answer is $12\times$ more accurate. This means the step-function model of the soliton boundary is inadequate for the CD. The CD vortex's geometric overlaps with the gauge configurations contribute to the transformation law at all scales, not just above M_VL.

Proposed revision to Table R.3: The row "Frozen vortex (below threshold): Particle too heavy to excite → contributes nothing" should be revised to:

"Frozen vortex (below threshold): Too heavy to excite as a real propagating state. Its geometric overlaps with the gauge configurations persist and may contribute to the transformation law. The step-function approximation (contribution = 0 below threshold) is adequate when the vortex's beta shifts are small relative to the SM betas. For vortexes with large beta shifts (e.g., the CD with $\Delta b_2 = 1$, which is 31% of $|b_2\_SM|$), the step-function approximation may fail, and the full geometric contribution at all scales may be required."

A6. From QED to GR — The Consistent Picture

The series has traced integer rules across domains. Here is how the vortex picture applies consistently from the lowest to the highest energies.

QED (PHYS-5). The electromagnetic coupling α_EM runs because charged vortexes (electron, muon, quarks) overlap geometrically with the U(1) gauge configuration. Each vortex screens the bare charge by an amount determined by its charge squared Q^2 and its color multiplicity N_c — both Level 1 geometric quantities. The transformation law b_EM = −(4/3)ΣQ^2N_c counts the geometric overlaps. Every charged vortex in the configuration contributes. None is omitted because it is "too heavy" — in the full QED calculation, all charged vortexes contribute at all scales, with heavier ones contributing less (suppressed by their inertia) but never zero.

QCD (PHYS-6). The strong coupling α_s runs because colored vortexes (quarks) and the gluon vortex configuration itself (self-interaction) overlap geometrically. The gauge self-coupling (−11 from the SU(3) adjoint) antiscreens — the gluon vortex configuration makes the force stronger at low energy. Quark vortexes screen (+2/3 per flavor). The balance −11 + (2/3)n_f determines asymptotic freedom. The confinement wall at Λ _QCD (~0.3 GeV) is where α_s approaches O(1) and the perturbative integer rules break down — the vortex interactions become too strong for the weak-coupling approximation.

Electroweak (PHYS-12). The three gauge configurations SU(3) $\times$ SU(2) $\times$ U(1) have distinct vortex structures with distinct geometric overlaps to matter vortexes. The SM betas $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$ count these overlaps. Each integer traces to a specific geometric relationship: the 41 in b_1 comes from the sum of $Y^2 \times$ dim contributions over all SM vortexes. The 19 in b_2 comes from the SU(2) adjoint (22/3) minus the quark and lepton doublet screening (4) minus the Higgs doublet (1/6). The 7 in b_3 comes from the SU(3) adjoint (11) minus three generations of quark screening (4). All Level 1 from the geometry.

Unification (PHYS-13, PHYS-26–30). The three gauge couplings, running according to their respective integer rules, converge at high energy. The Cabibbo Doublet — a vortex in the (3,2,1/6) configuration — modifies the integer rules by (Δb_1 = 1/15, Δb_2 = 1, Δb_3 = 1/3). These shifts come from the CD vortex's geometric overlaps with all three gauge configurations simultaneously. The CD participates in SU(3) as a triplet (S_2 = 1/2), in SU(2) as a doublet (S_2 = 1/2), and in U(1) through its hypercharge (Y = 1/6). The modified betas produce near-exact unification: α_s predicted to 0.33%, $\sin^2\theta$_W to 0.048%.

The no-threshold result (this paper). The CD vortex's geometric overlaps do not switch off at M_VL. The no-threshold configuration — which includes the CD's geometric contribution at all scales — gives 12 $\times$ better predictions than the threshold configuration. This is consistent with the vortex picture: the geometric overlaps are properties of the vortex topology, not of the available energy. The CD is a triplet-doublet-hypercharged vortex at every scale. Its contribution to the integer rules should reflect this.

Gravity. The series has not yet computed gravitational effects in the vortex framework. But the principle extends: mass is inertia (R3), inertia is the energy cost of redistributing a vortex configuration, and every vortex with inertia gravitates. The same vortex that contributes to gauge coupling running through its geometric overlaps also contributes to the gravitational configuration through its inertia. The coupling between gauge geometry and gravitational geometry is through the vortex itself — the same entity participates in both. The QED-to-GR bridge, if it exists, runs through the vortex: the geometric overlaps determine the gauge running (integer rules), and the inertia of the vortex determines the gravitational contribution (mass-energy). Both are properties of the same circulating energy configuration.

A7. The Position Correction

The paper body frames the no-threshold result as a "puzzle" and an "inversion" of the decoupling theorem. From the vortex framework, it is neither.

The puzzle framing assumes the step-function threshold is the expected answer. It is not. It is an approximation that works when a vortex's geometric contribution is small enough to round to zero below its inertia scale. For the CD, the contribution is not small (Δb_2 = 1, 31% of $|b_2$_SM$|$), and the approximation fails.

The inversion framing assumes the decoupling theorem predicts the CD should not contribute below M_VL. The theorem predicts the contribution is SUPPRESSED, not ABSENT. The step function rounds "suppressed" to "zero." The no-threshold configuration rounds "suppressed" to "full." The data says "full" is 12 $\times$ closer to reality than "zero." The truth is between them, but much closer to "full."

The corrected framing: the CD is a vortex with geometric overlaps (3,2,1/6) to all three gauge configurations. These overlaps produce definite integer rule changes (Δb_1 = 1/15, Δb_2 = 1, Δb_3 = 1/3). The overlaps exist at all scales because they are topological properties of the vortex, not energy-dependent thresholds. The no-threshold result confirms that the integer rules should reflect the full vortex configuration at all scales. The step-function approximation, which erases the CD's geometric contribution below its inertia scale, introduces a 4% error that is 12 $\times$ larger than the 0.33% error of the full-configuration treatment.

This is not a puzzle. It is the vortex framework working as expected.

A8. Revision to Table R.3

The following revision is proposed for Table R.3, row "Frozen vortex":

Current: "Frozen vortex (below threshold): Particle too heavy to excite. Contributes nothing."

Revised: "Frozen vortex (below threshold): Too heavy to excite as a propagating state. Geometric overlaps with gauge configurations persist at all scales. The step-function approximation (contribution = 0 below inertia scale) is adequate for vortexes with small beta shifts. For vortexes with large beta shifts (Δb comparable to SM b), the full geometric contribution at all scales may be required. PHYS-35 demonstrates this for the CD: the full-configuration treatment (no threshold) outperforms the step-function treatment by 12 $\times$."

This addendum restates the PHYS-35 finding in series language. The no-threshold result is not a puzzle — it is the expected behavior of a vortex whose geometric overlaps with the *gauge configurations exist at all scales. The step-function threshold is an approximation that erases this geometry below the inertia scale. For the CD, this erasure introduces a 12 × error. The vortex framework (R3, R4, R5) predicts this: mass is inertia, not a boundary of existence; the vortex exists at all scales; its geometric contribution to the integer rules reflects its topology, not the available energy.*

[**HOWL-PHYS-35-2026**] The No-Threshold Puzzle — More CD Running Means Better Predictions. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-35-2026

[**HOWL-PHYS-1-2026**] The Inertial Vortex. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOW
PHYS-1-2026

[**HOWL-PHYS-13-2026**] Gauge Coupling Unification and Minimal BSM Content. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-13-2026

[**HOWL-PHYS-27-2026**] $\sin^2\theta$_W from 3/8 — The Weak Mixing Angle as a Running Prediction. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-27-2026

[**HOWL-PHYS-30-2026**] α_s from Unification — The Strong Coupling as a Derived Quantity. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-30-2026